

EXPERIMENT 21 to 23

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21. Write a Pandas program to swap the cases of a specified character column in a given DataFrame.

CODE:

```
import pandas as pd
```

```
# Create a DataFrame
```

```
df = pd.DataFrame({  
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],  
    'City': ['Chennai', 'Delhi', 'Mumbai', 'Kolkata']  
})
```

```
print("Original DataFrame:")  
print(df)
```

```
# Swap the case of the Name column
```

```
df['Name'] = df['Name'].str.swapcase()
```

```
print("\nDataFrame after swapping case:")
```

```
print(df)
```

OUTPUT:

```
>>> ----- KEDJAKI -----  
>>>  
Original Data:  
Name      City  
Alice      Chennai  
Bob         Delhi  
Charlie      Mumbai  
David      Kolkata  
  
After swapping case:  
Name      City  
aLiCe      Chennai  
bOb         Delhi  
cHARLIE      Mumbai  
dAVID      Kolkata  
>>>
```

22. Write a Python program to draw a line with suitable label in the x axis, y axis and a title.

CODE:

```
import matplotlib.pyplot as plt
```

```
# Data
```

```
x = [0, 10, 20, 30, 40, 50]
```

```
y = [0, 30, 60, 90, 120, 150]
```

```
# Draw line
```

```
plt.plot(x, y)
```

```
# Add labels and title
```

```
plt.xlabel("X - axis")
```

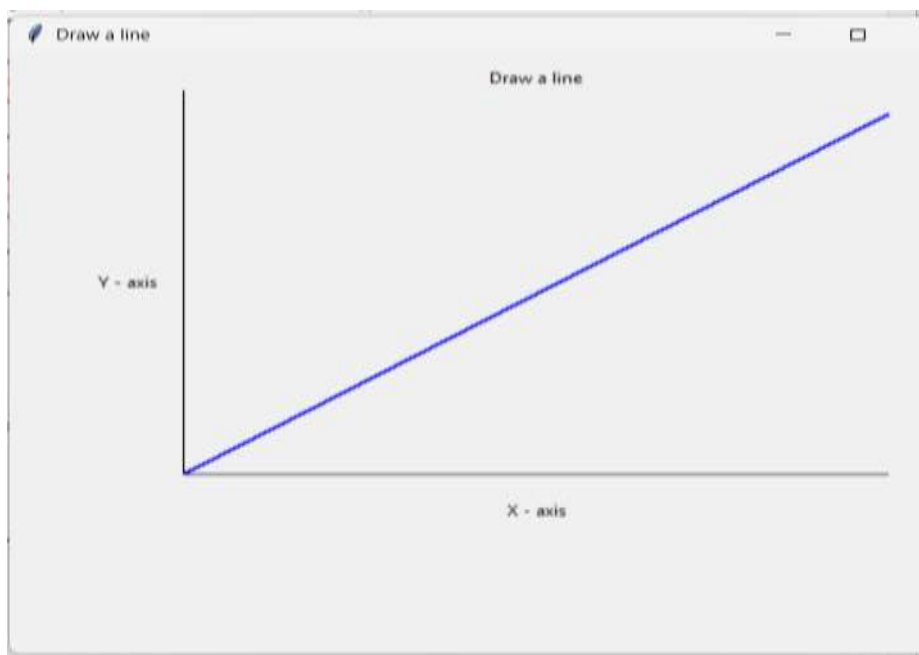
```
plt.ylabel("Y - axis")
```

```
plt.title("Draw a line")
```

```
# Display graph
```

```
plt.show()
```

OUTPUT:



23. Write a Python program to draw a line using given axis values taken from a text file, with suitable label in the x axis, y axis and a title.

Test Data:

test.txt

1 2

2 4

3 1

CODE:

```
from tkinter import *

# Create test.txt automatically
file = open("test.txt", "w")
file.write("1 2\n")
file.write("2 4\n")
file.write("3 1\n")
file.close()

# Read values from text file
x = []
y = []

file = open("test.txt", "r")

for line in file:
    a, b = line.split()
    x.append(int(a))
    y.append(int(b))

file.close()

# Create graph window
```

```
root = Tk()
root.title("Sample graph")

canvas = Canvas(root, width=600, height=500)
canvas.pack()

# Graph area
left = 80
bottom = 400
right = 550
top = 50

# Draw axes
canvas.create_line(left, bottom, right, bottom)
canvas.create_line(left, bottom, left, top)

# Title and labels
canvas.create_text(315, 25, text="Sample graph!")
canvas.create_text(315, 450, text="X - axis")
canvas.create_text(30, 225, text="Y - axis")

# Draw line using values from file
points = []

for i in range(len(x)):
    px = left + (x[i] - 1) * 200
    py = bottom - (y[i] - 1) * 100
    points.append((px, py))

for i in range(len(points) - 1):
    canvas.create_line(
```

```
points[i][0], points[i][1],  
points[i+1][0], points[i+1][1],  
fill="blue",  
width=2  
)  
root.mainloop()
```

OUTPUT:

